RoboCup 2023 Main Control Board Prototype

Project Description

The **RoboCup 2023 Main Control Board Prototype** is a custom-designed embedded control system developed for robotic applications participating in the RoboCup competition. The board serves as the central interface between a **Raspberry Pi** and an **Arduino Mega 2560 Pro**, enabling efficient communication between high-level processing and low-level motor control.

The **Raspberry Pi** is responsible for computationally intensive tasks such as sensor data acquisition, image processing, computer vision, decision-making, and autonomous navigation. The **Arduino Mega 2560 Pro** manages real-time motor control, actuator operation, and peripheral interfacing to ensure precise and responsive robot movement.

The custom PCB integrates power distribution, communication interfaces, and connector routing into a compact and reliable design. By consolidating all major system connections onto a single board, the prototype improves wiring organization, enhances system reliability, and simplifies robot assembly and maintenance.

Hardware Components

Component	Quantity	Purpose
Raspberry Pi	1	High-level processing, sensor management, and autonomous decision-making
Arduino Mega 2560 Pro	1	Real-time motor control and peripheral management
Custom PCB	1	Central interface for system integration
Motor Drivers	As Required	Controls DC or brushless motors
Sensors	Multiple	Environmental sensing and robot navigation
Communication Interfaces	Multiple	Data exchange between Raspberry Pi and Arduino
Power Distribution Circuit	1	Provides regulated power to all system components
Connectors & Headers	Multiple	Secure connections for modules and peripherals

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Key Features

- Centralized hardware interface for RoboCup robots
- Seamless communication between Raspberry Pi and Arduino Mega 2560 Pro

- Dedicated motor control architecture
- Support for multiple sensors and actuators
- Integrated power distribution and signal routing
- Custom PCB for compact and reliable system integration
- Modular design for easy maintenance and future expansion
- Optimized for autonomous robotics applications

Applications

- RoboCup competition robots
- Autonomous mobile robots
- Educational robotics platforms
- Research and development in robotics
- Multi-controller embedded systems
- Robot navigation and motion control
- Custom robotic automation projects